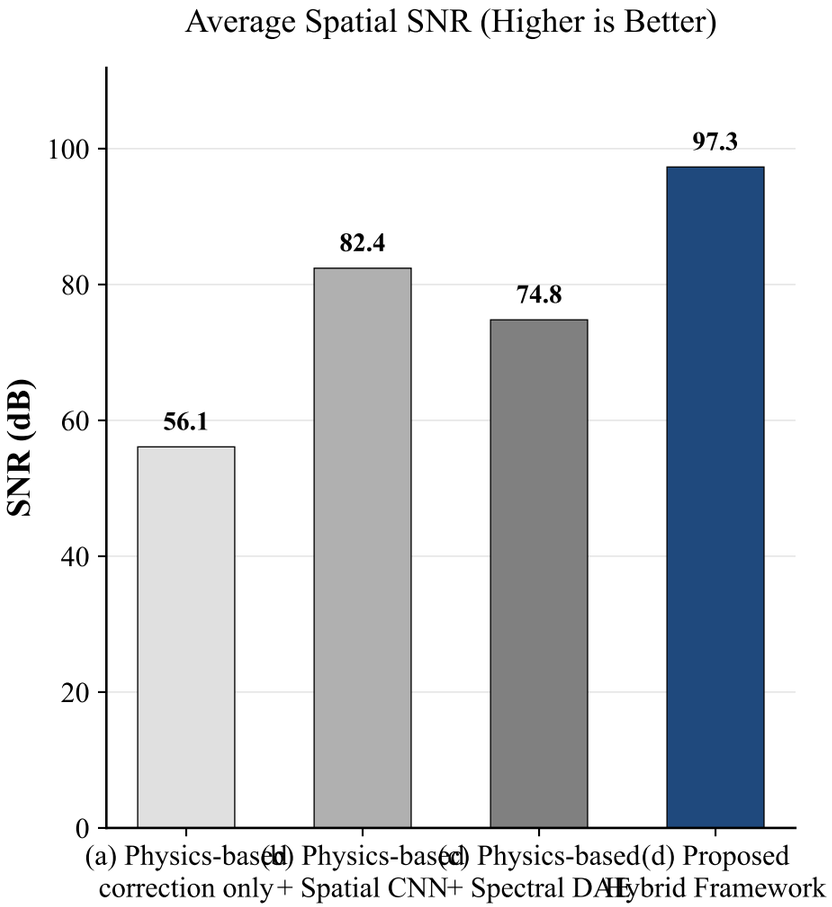
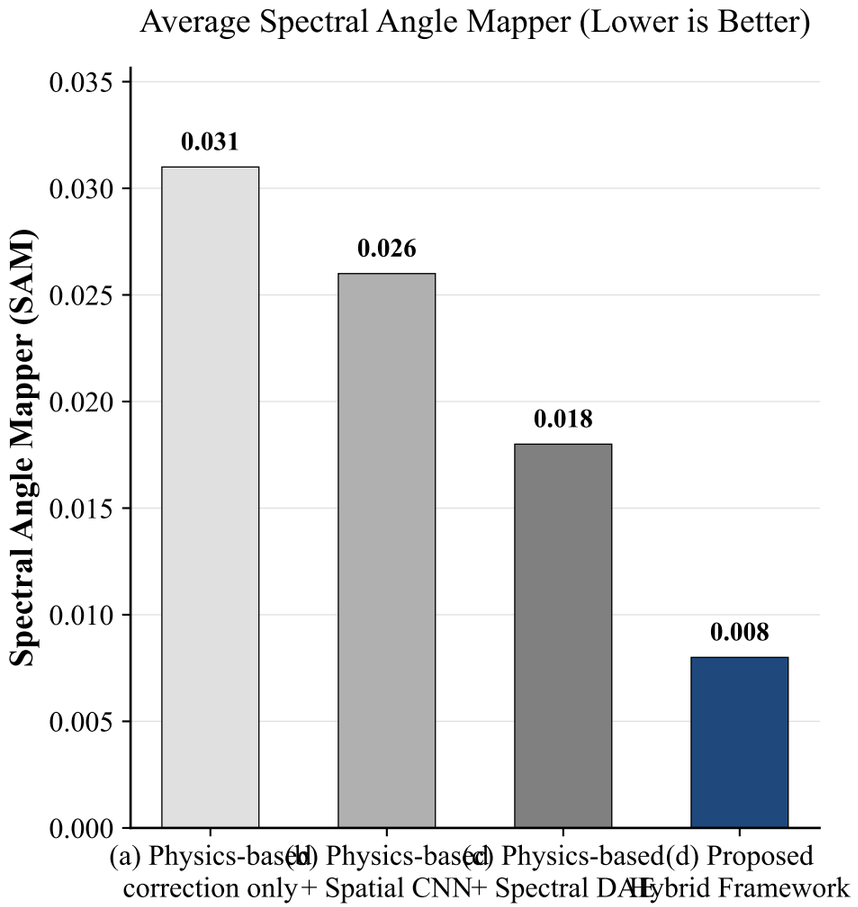
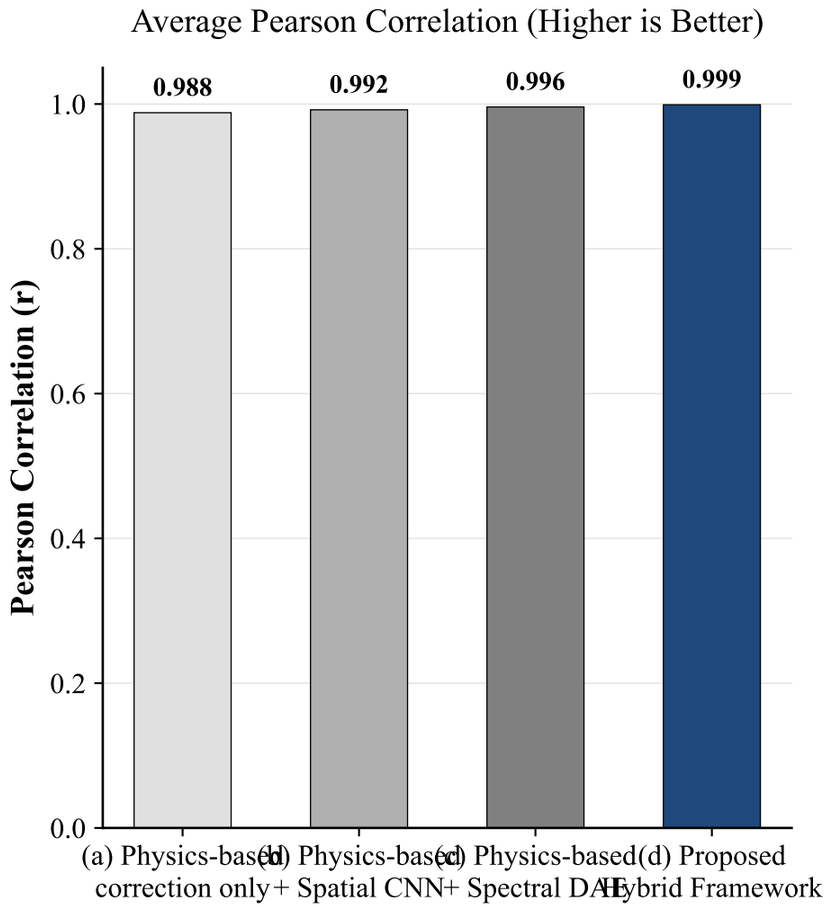
Ablation Study of the Proposed Hybrid Denoising Framework







Key Observations

- Physics-based correction provides the baseline removal of deterministic sensor artefacts.
- Adding the Spatial CNN substantially improves spatial restoration by suppressing residual stripe noise.
- Adding the Spectral Noise2Void DAE better preserves spectral signatures while reducing random Gaussian noise.
- Combining both modules produces the best overall balance between spatial enhancement and spectral fidelity.